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A spatial and seasonal assessment of river water chemistry across North West England

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ABSTRACT

This paper presents information on the spatial and seasonal patterns of river water chemistry at approximately 800 sites in North West England based on data from the Environment Agency regional monitoring programme. Within a GIS framework, the linkages between average water chemistry (pH, sulphate, base cations, nutrients and metals) catchment characteristics (topography, land cover, soil hydrology, base flow index and geology), rainfall, deposition chemistry and geo-spatial information on discharge consents (point sources) are examined. Water quality maps reveal that there is a clear distinction between the uplands and lowlands. Upland waters are acidic and have low concentrations of base cations, explained by background geological sources and land cover. Localised high concentrations of metals occur in areas of the Cumbrian Fells which are subjected to mining effluent inputs. Nutrient concentrations are low in the uplands with the exception sites receiving effluent inputs from rural point sources. In the lowlands, both past and present human activities have a major impact on river water chemistry, especially in the urban and industrial heartlands of Greater Manchester, south Lancashire and Merseyside. Over 40% of the sites have average orthophosphate concentrations >0.1 mg-P l^{-1}. Results suggest that the dominant control on orthophosphate concentrations is point source contributions from sewage effluent inputs. Diffuse agricultural sources are also important, although this influence is masked by the impact of point sources. Average nitrate concentrations are linked to the coverage of arable land, although sewage effluent inputs have a significant effect on nitrate concentrations. Metal concentrations in the lowlands are linked to diffuse and point sources. The study demonstrates that point sources, as well as diffuse sources, need to be considered when targeting measures for the effective reduction in river nutrient concentrations. This issue is clearly important with regards to the European Union Water Framework Directive, eutrophication and river water quality.
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1. Introduction

River water chemistry is determined by the interactions between soils and underlying geology, topographic features, rainfall and atmospheric inputs, anthropogenic sources (e.g. agricultural runoff and effluent discharges) and catchment hydrology (Billett et al., 1996; Robson and Neal, 1997; Smart et al., 1998; Cooper et al., 2000; Jarvie et al., 2002; Daly et al., 2002; Buck et al., 2004). Within a Geographical Information System (GIS) framework, spatial datasets, such as land use and geology, have been used to explore the controls on river water quality in a variety of environments and across a range of scales. The catchment characteristic approach has been used within the context of surface water acidification (e.g. Kernan et al., 1998; Thornton and Dise, 1998; Cooper et al., 2004; Cresser et al., 2006; Evans et al., 2006), nutrient transfers through river basins (Ekholm et al., 2000; Ferrier et al., 2001; Daly et al., 2002; Donohue et al., 2005; Jarvie et al., 2002; Davies and Neal, 2004, 2007; Evans et al., 2006; Meynendonckx et al., 2006; Helliwell et al., 2007) and sediment dynamics in fluvial systems (e.g. Siakeu et al., 2004; Takagi et al., 2007; Hashimoto et al., 2008).

Understanding the relationships between catchment characteristics and river water chemistry provides a base for determining how future changes in land use and climate will impact on river water quality and functioning (Robson and Neal, 1997; Jarvie et al., 2002). Therefore, it is important to determine the processes that regulate stream water chemistry in landscapes under increasing pressure from human population, whether from urbanisation or more intensive food production. This understanding will help underpin environmental management, especially in the context of meeting the hydrochemical and ecological goals of the European Union Water Framework Directive (WFD) (European Community, 2000).

Within the UK, considerable amounts of water quality data have been and are continuing to be collected by environmental regulatory bodies, such as the Environment Agency of England and Wales and the

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Scottish Environment Protection Agency. Water quality datasets held by these bodies have been used to explore controls on stream water chemistry for some major UK river basins, such as the Humber and the Tweed (e.g. Neal et al., 1997; Robson and Neal, 1997; Jarvie et al., 2000; Oguchi et al., 2000; Jarvie et al., 2002). However, there is paucity of information on the variability in river water chemistry in relation to catchment characteristics for many UK rivers.

Here we examine one critical area based on the industrial and urban conurbations/heartlands of the North West of England and the major rivers draining this area entering the Irish Sea. The rivers of this region are subjected to numerous pressures including point and diffuse pollution. The importance of this area is recognised for example with the River Ribble which provided a key typology for a European Union river basin network designed to test the WFD implementation guidelines (Fox et al., 2004; EA, 2009a). This paper builds upon the water quality mapping approaches used by Robson and Neal (1997) and Oguchi et al. (2000) to explore the spatial distribution of river water chemistry across the North West region using Environment Agency water quality data. The methods employed by Thornton and Dise (1998), Jarvie et al. (2002) and Davies and Neal (2004, 2007) are used to explore empirical relationships between a wide range of catchment characteristics and average river water chemistry (acidity and base cations, nutrients and metals). This is extended by GIS-based analysis of geo-spatial information on Environment Agency consented discharges from point sources and examination of seasonal water chemistry data. We address three key questions in this paper:

1. Are there regional patterns in river water chemistry?
2. Are there linkages between river water chemistry and catchment characteristics?
3. What are the major sources of acidity, base cations, nutrients and metals?

2. Study area

The North West of England (Fig. 1A) is one of the UK's most varied regions, with the rural upland landscapes of the north, to the populated lowland centres of the south (Fig. 1B). The region has a variety of land use types including intensive agriculture on the Lancashire plain, open moorland in Cumbria and heavily urbanised and industrial areas of

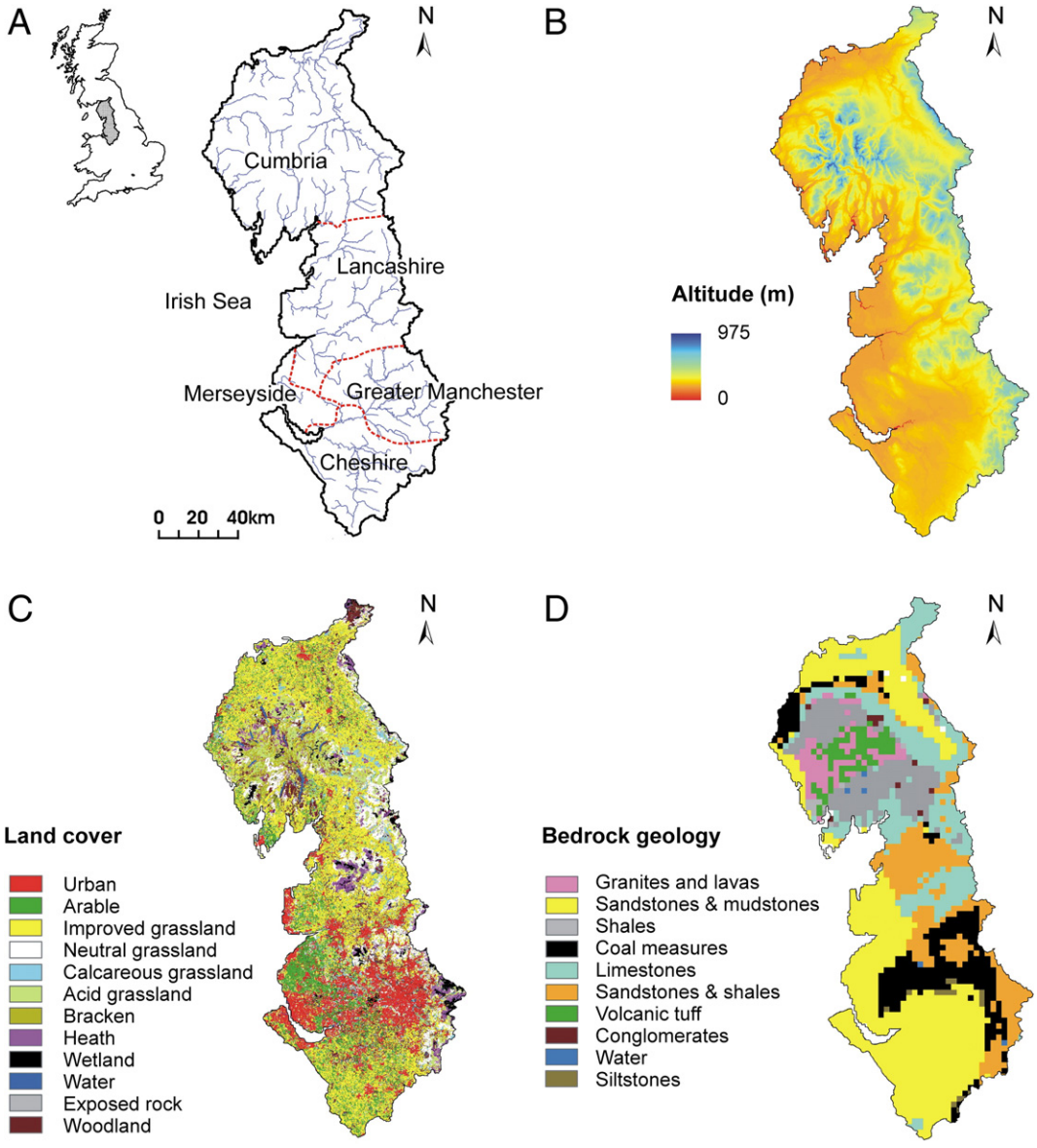


Fig. 1. North West study area showing major rivers (A); altitude (B); land cover (C); bedrock geology (D).

Manchester and Liverpool (Fig. 1C). Major rivers include the Eden and Duddon in Cumbria, the Lune, Ribble and Wyre in Lancashire and the Irwell and Mersey in Greater Manchester. Many of the rivers in Manchester and Liverpool have been heavily modified. Industrial activity in the region over the last few hundred years resulted in severe pollution of the many rivers in the south of the region, particularly those in the Mersey basin (DoE, 1982; Nolan and Guthrie, 1998; Jones, 2000; Burton, 2003). Although there has been significant improvements in regional water quality in recent years (Harland et al., 2000; Mersey Basin Campaign, 2009), many of the North West's rivers are still subjected to significant sewage and industrial inputs and/or agricultural runoff (Burton, 2003; George et al. 2004; Barker et al., 2005; Davies and Neal, 2007). Although there has been a reduction in the atmospheric deposition of pollutant sulphur across much of the North West, upland areas in Cumbria and along the Pennine fringe still receive substantial levels of nitrogen deposition (Tipping et al., 2008). Surface water acidification is still a problem in some fluvial systems in the region (Evans and Jenkins, 2000; Curtis et al., 2005; Davies et al., 2005; Daniels et al., 2008; Tipping et al., 2008).

3. Methods

3.1. Water chemistry data

The water chemistry dataset used in this investigation was supplied by the Environment Agency (EA) of England and Wales. Data were made available for Cumbria, Lancashire, Merseyside and Greater Manchester and north Cheshire (Fig. 1). Determinands chosen were pH, calcium (Ca^{2+}), magnesium (Mg^{2+}), sulphate (SO_4^{2-}), ammonium (NH_4^+-N), nitrite (NO_2^--N), nitrate (NO_3^--N), orthophosphate (PO_4^{3-}-P), iron (Fe^{3+}), lead (Pb^{2+}), copper (Cu^{2+}), zinc (Zn^{2+}). Measurements made by the EA are on unfiltered water samples. To ensure harmonisation with spatial data (see below), good temporal coverage, and to remove bias from monitoring sites measured only sporadically (e.g. accidental discharge events), a series of site selection criteria were used. Only river monitoring data for the period 1995 to 2001 (inclusive) were used in this study. Monitoring sites were only selected where more than 60 readings were available for the entire period. Tidal reaches of rivers were excluded. Hydrological processes play a major role in determining the dynamic nature of stream water chemistry (Neal et al., 1997). Due to the extensive number of sites in this investigation, flow data was not considered. Instead, seasonal water quality data (average determinand concentrations for summer and winter months) were used as a surrogate for low flow and high flow conditions. Sites were selected where at least 15 readings were available for summer months (June, July and August) and at least 15 readings for winter months (December, January and February). From a total of 1395 EA monitoring sites across the North West, the number of sites for each determinand was: 798 (pH, NH_4^+-N, NO_2^--N, NO_3^--N and PO_4^{3-}-P), 420 (Zn^{2+}), 413 (Ca^{2+}), 385 (Mg^{2+}), 195 (SO_4^{2-}), 76 (Fe^{3+}), 62 (Pb^{2+}), and 60 (Cu^{2+}). For each site the average concentration of each determinand was calculated for the entire period, together with summer and winter average concentrations. Robson and Neal (1997) highlight the importance of examining time series when averaging river water chemical data due to the potential for time variations in determinand detection limits (especially metals). Since the period 1995 to 2001 was used in this analysis, it is assumed that there are no significant step changes in analytical methodology.

3.2. Spatial data

A variety of spatial datasets were used in this investigation. Many of these datasets were provided by the Centre for Ecology and Hydrology (CEH). The spatial datasets are described in detail below.

3.2.1. CEH Wallingford Digital Terrain Model (DTM)

Available as a 50 m (horizontal) and 0.1 m (vertical) resolution grid (Morris and Flavin, 1990). The DTM for the North West region is shown in Fig. 1B.

3.2.2. CEH Land Cover Map 2000 (LCM2000)

Available as a 25 m UK-wide grid. The 26 land cover classes available in the LCM2000 were aggregated into 12 classes: arable, bracken, exposed rock, grassland (acid, calcareous, improved and neutral), heathland, urban, water, wetland and woodland. Fig. 1C shows the land cover types across the North West.

3.2.3. CEH Hydrology of Soil Types (HOST)

Available as a 1 km UK-wide grid. The 29 HOST classes were simplified into 4 classes: permeable, impermeable (thin), gleyed and peat soils. See Boorman et al. (1995) for a description of HOST.

3.2.4. Base Flow Index (BFI)

Available as a 1 km UK-wide grid of BFI values. BFI values range between 0 and 1 and indicate the ratio of base flow to total flow volume (Gustard et al., 1992).

3.2.5. British Geological Survey Bedrock Geology and Superficial Deposits

Available as 1:625,000 scale data via the British Geological Survey (BGS) Digital Geological Map of Great Britain (DiGMapGB). There are over 100 bedrock classes, of which approximately 30 occur within the North West region. These were aggregated to form 9 new classes: coal measures, conglomerates, granites and lavas, limestones, shales, siltstones, sandstones and shales, sandstones and mudstones, and volcanic tuff. These are shown in Fig. 1D. There are five major superficial classes in the North West region. Aeolian sand and glacial sand were aggregated into one sand class. The other classes were alluvium, peat, till and 'no superficial deposits'.

3.2.6. Meterological Office Standard Annual Average Rainfall (SAAR)

Available as a 1 km grid based on the average annual rainfall for the period 1961–1990.

3.2.7. CEH atmospheric deposition

Data available as a 5 km UK-wide grid of NH_3/NH_4, non marine Ca + Mg, non marine SO_2/SO_4 and $NO_2/NO_3/HNO_3$ for the period 1998 to 2000. See Smith et al. (2000) for a description of the atmospheric deposition model.

3.2.8. Environment Agency discharge consents

These represent the location and type of EA discharge consents or 'consent to discharge'. The EA issue these consents for the discharge of domestic and industrial effluent into surface waters. Therefore, this dataset provides an indication of known point sources within the study region. There are approximately 20 types of discharge consents, but 6 of the most common categories were selected. The type and number of consents in the study region are: crude sewage effluent (649), treated sewage effluent (3748), storm overflow sewage effluent (2140), crude trade effluent (622), treated trade effluent (1092) and trade mine-water (123).

3.3. Spatial analysis

All spatial datasets were loaded into the Terrain Analysis System (TAS), a freely available software package for performing spatial analysis operations for environmental applications (Lindsay, 2005). For the North West region, flow direction, flow accumulation and the stream network were derived using the DTM. Prior to delineation of catchment boundaries for the EA sites, the outlet points were automatically repositioned onto the digital stream network using the advanced outlet repositioning approach (AORA) (Lindsay et al., 2008).

For each of the delineated catchments the average elevation (m), slope (°), aspect (°), BFI, rainfall (mm) and atmospheric deposition (keq $ha^{-1}y^{-1}$) were derived using the raster calculator function in TAS. Catchment area and the percentage of total catchment area covered by each of the land cover, HOST, bedrock and superficial classes were also derived.

For each catchment, the total number of the six types of discharge consents was calculated, thus enabling consent density (in km^{-2}) to be derived. The proximity (in m) of each EA monitoring site to the nearest upstream discharge consent was also calculated using a simple program written in Microsoft Visual Basic. Three density categories (low; <0.1 km^{-2}, moderate; 0.1 to 0.3 km^{-2}, and high; >0.3 km^{-2}) and two proximity categories (close; <1000 m and distant; >1000 m) were used to group EA monitoring sites. These discharge consent groups were: none, low density/distant, low density/close proximity, moderate density/distant, moderate density/close proximity, high density/distant and high density/close proximity. Although the selection of the thresholds for density and proximity categories is qualitative, it does provide a relative measure of the impact of point sources within each catchment.

3.4. Water chemistry maps and statistical techniques

The distributions of average determinand concentrations were mapped using ArcGIS 9.2. The concentration values were grouped into five categories using the natural break method. This is a common method for categorising environmental data and has been used in similar studies (e.g. Oguchi et al., 2000). As well as mapping average determinand concentrations, the main river basin districts within the North West region were mapped using the Environment Agency scheme. Inter-determinand relationships were assessed using bivariate Pearson correlation. The relationships between average determinand concentrations were also visually examined in a correlation matrix plot. The degree of associations between water chemistry determinands, catchment characteristics, rainfall, deposition chemistry and discharge consent metrics was also measured using bivariate Pearson correlation. Previous research has shown quantitative linkages between catchment land cover (particularly the proportion of arable land and catchment urbanisation) and river nutrient concentrations (e.g. Jarvie et al., 2002; Davies and Neal, 2007). To explore these linkages, scatter plots were made for selected nutrient determinands and potential key environmental drivers. All water quality data manipulation and queries were performed in Microsoft Access. SPSS v16 was used for statistical analyses.

4. Results

4.1. River monitoring and discharge consents

Fig. 2A shows the 798 water quality monitoring sites across the North West region which met the site selection criteria. There is good spatial coverage of monitoring sites. Fig. 2B shows the spatial distribution of discharge consents. The highest density of discharge consents is in Greater Manchester, Merseyside and south Lancashire while there is a paucity of discharge consents for Cumbria (Fig. 2B).

4.2. Patterns in average water chemistry

The patterns of average water chemistry across the North West are presented as a series of maps in relation to acidity and base cations

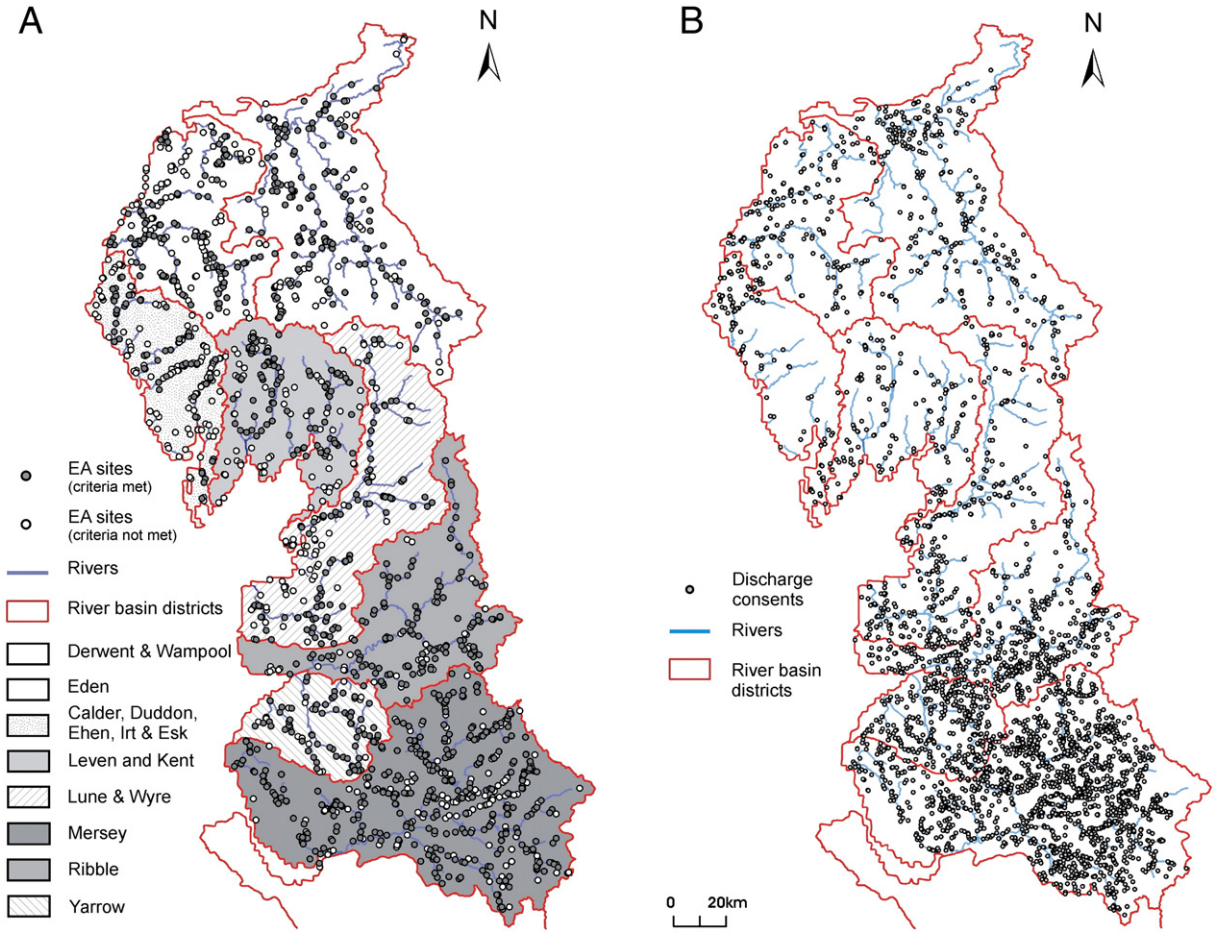


Fig. 2. Location of Environment Agency river monitoring sites (A) and Environment Agency discharge consent sites (B) in the North West.

(pH, Ca^{2+}, Mg^{2+} and SO_4^{2-}: Fig. 3A, B, C, D, respectively), nutrients (NH_4^+-N, NO_2^--N, NO_3^--N and PO_4^{3-}-P: Fig. 4A, B, C, D, respectively) and metals (Fe^{3+}, Pb^{2+}, Cu^{2+} and Zn^{2+}: Fig. 5A, B, C, D, respectively). The most common feature of the maps is a distinction between the highly populated and industrial areas of Greater Manchester and Merseyside and the agricultural and upland areas of Lancashire and Cumbria.

4.2.1. Acidity and base cations

pH values are fairly uniform across the region, with approximately 65% of sites having an average pH of between 7.4 and 8. Exceptions to this are the lower values in rivers draining the Cumbrian and Bowland fells (pH 5.74 to 6.55), the Pennine Moors to the north-east of Manchester (pH 4.47 to 6.06) and the Southern Pennines to the east of Manchester (pH 6.12 to 6.75). Lower pH values also occur in streams draining the lowland wet and dry heath in the Mersey basin (pH 6.99 to 7.21; Fig. 3A). High Ca^{2+} concentrations (>100 mg l^{-1}) occur in the southern part of the region, especially in the lowland reaches of the Mersey, Ribble, Lune, Wyre and Yarrow basins as well as along the Eden and coastal plain of the Derwent basin. Low Ca^{2+} concentrations (<10 mg l^{-1}) generally occur in the upland areas of the region (Fig. 3B). The spatial pattern of Mg^{2+} (Fig. 3C) has a similar distribution to that of Ca^{2+} with a concentration range of 0.001 to 120 mg l^{-1}. The SO_4^{2-} data are too sparse to identify any regional patterns, but relatively high SO_4^{2-} concentrations of 100 to 485 mg l^{-1} generally occur in urban and industrial areas, with values <20 mg l^{-1} in the rural areas (Fig. 3D).

4.2.2. Nutrients

Elevated NH_4^+ and NO_2^- concentrations (5.70 to 30 mg-N l^{-1}, Fig. 4A, and 0.30 to 1.60 mg-N l^{-1}, Fig. 4B, respectively) occur in the heavily urbanised and industrial areas of south Lancashire, Greater Manchester and Merseyside. High to intermediate concentrations also occur in some of the Cumbrian towns, such as Penrith. Low NH_4^+ and NO_2^- concentrations occur in the rural and upland areas (0.10 to 0.50 mg-N l^{-1}, Fig. 4A, and 0.001 to 0.05 mg-N l^{-1}, Fig. 4B respectively). High NO_3^- concentrations

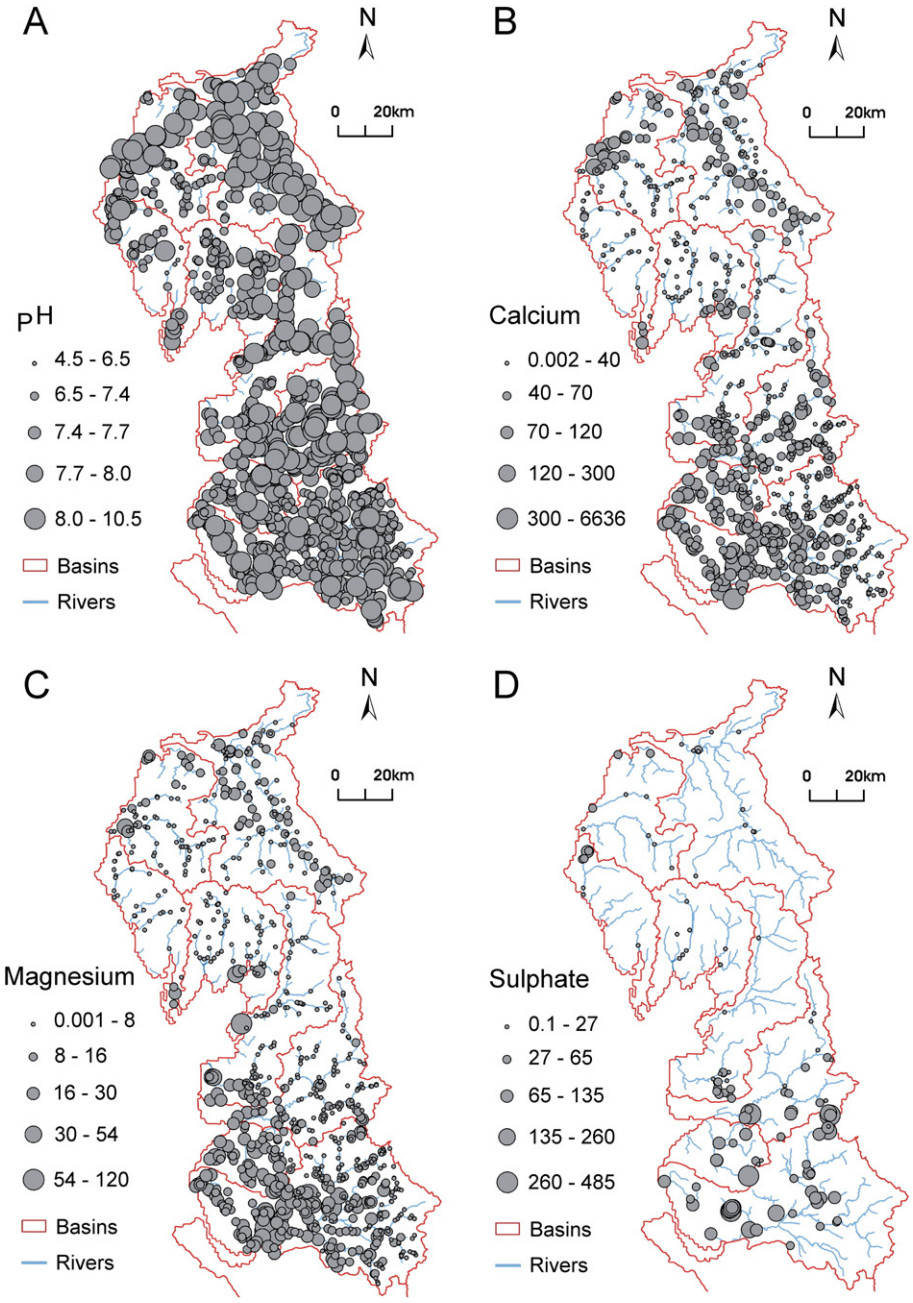


Fig. 3. Maps of average water chemistry across the North West. (A) pH; (B) Ca^{2+} (mg l^{-1}); (C) Mg^{2+} (mg l^{-1}); (D) SO_4^{2-} (mg l^{-1}).

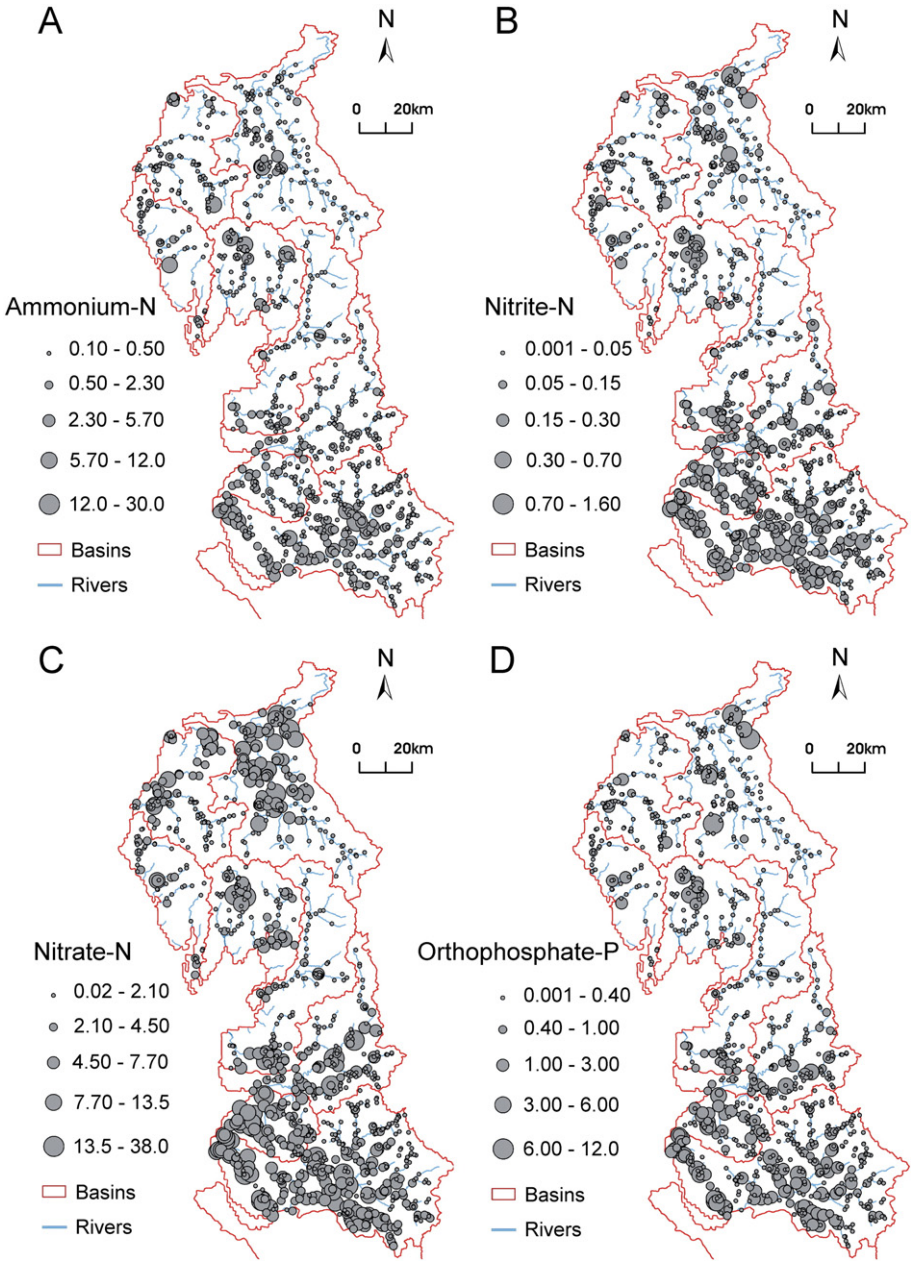


Fig. 4. Maps of average water chemistry across the North West. (A) NH_4^+ (mg-N l^{-1}); (B) NO_2^- (mg-N l^{-1}); (C) NO_3^- (mg-N l^{-1}); (D) PO_4^{3-} (mg-P l^{-1}).

are found in the lowland reaches of the Mersey basin, along the Ribble and Wyre, and the Sefton coast in Lancashire (7.70 to 38.0 mg-N l^{-1}, Fig. 4C). In Cumbria, high NO_3^- concentrations are found in the Leven and Kent basins, along the west coast plain and in the Eden basin (Fig. 4C). The distribution of PO_4^{3-} concentrations (Fig. 4D) shows some similarity to that of NO_3^--N, although elevated PO_4^{3-} concentrations in Cumbria only occur in a few locations. The PO_4^{3-} concentrations vary between 0.03 to 7.01 mg-P l^{-1} for the urban areas, to 0.001 to 11.9 mg-P l^{-1} for the rural counterparts.

4.2.3. Metals

High to intermediate Fe^{3+}, Pb^{2+}, Cu^{2+} and Zn^{2+} values are associated with urban and industrial areas in the north and south of the region (ranges 6000 to 43,000 μg l^{-1}, 30.0 to 345 μg l^{-1}, 35.0 to 145 μg l^{-1} and 100 to 525 μg l^{-1} respectively: Fig. 5). Although rural areas generally have lower Fe^{2+}, Pb^{2+}, Cu^{2+} and Zn^{2+} values (30.0 to 1000 μg l^{-1}, 0.40 to 5.00 μg l^{-1}, 0.50 to 6.00 μg l^{-1} and 5.00 to 20.0 μg l^{-1} respectively: Fig. 5) some rural streams in the Cumbrian Fells have high metal concentrations of up to 4263 μg-Fe l^{-1}, 40.7 μg-Pb l^{-1} and 354 μg-Zn l^{-1}.

4.3. Inter-determinand relationships

Table 1 shows correlation coefficients for average water chemistry determinand pairs. For the acidity-related and base cation determinands, there are statistically significant ($p<0.01$) positive correlations between Ca^{2+} and pH, Ca^{2+} and Mg^{2+}, Ca^{2+} and SO_4^{2-}, and Mg^{2+} and SO_4^{2-}. For nutrients, there are statistically significant ($p<0.01$) relationships between NH_4^+-N and NO_2^--N, NO_2^--N and NO_3^--N, NO_2^--N and PO_4^{3-}-P, NO_3^--N and Ca^{2+}, and NO_3^--N and PO_4^{3-}-P. For the metals, there are statistically significant ($p<0.01$) relationships between Pb^{2+} and Cu^{2+}, Zn^{2+} and Cu^{2+}, and Zn^{2+} and SO_4^{2-}.

Average SO_4^{2-} and PO_4^{3-}-P show a clear two way separation (Fig. 6A). One trend is characterised by high SO_4^{2-} but low PO_4^{3-}-P, and another trend is characterised by higher average PO_4^{3-}-P for a given

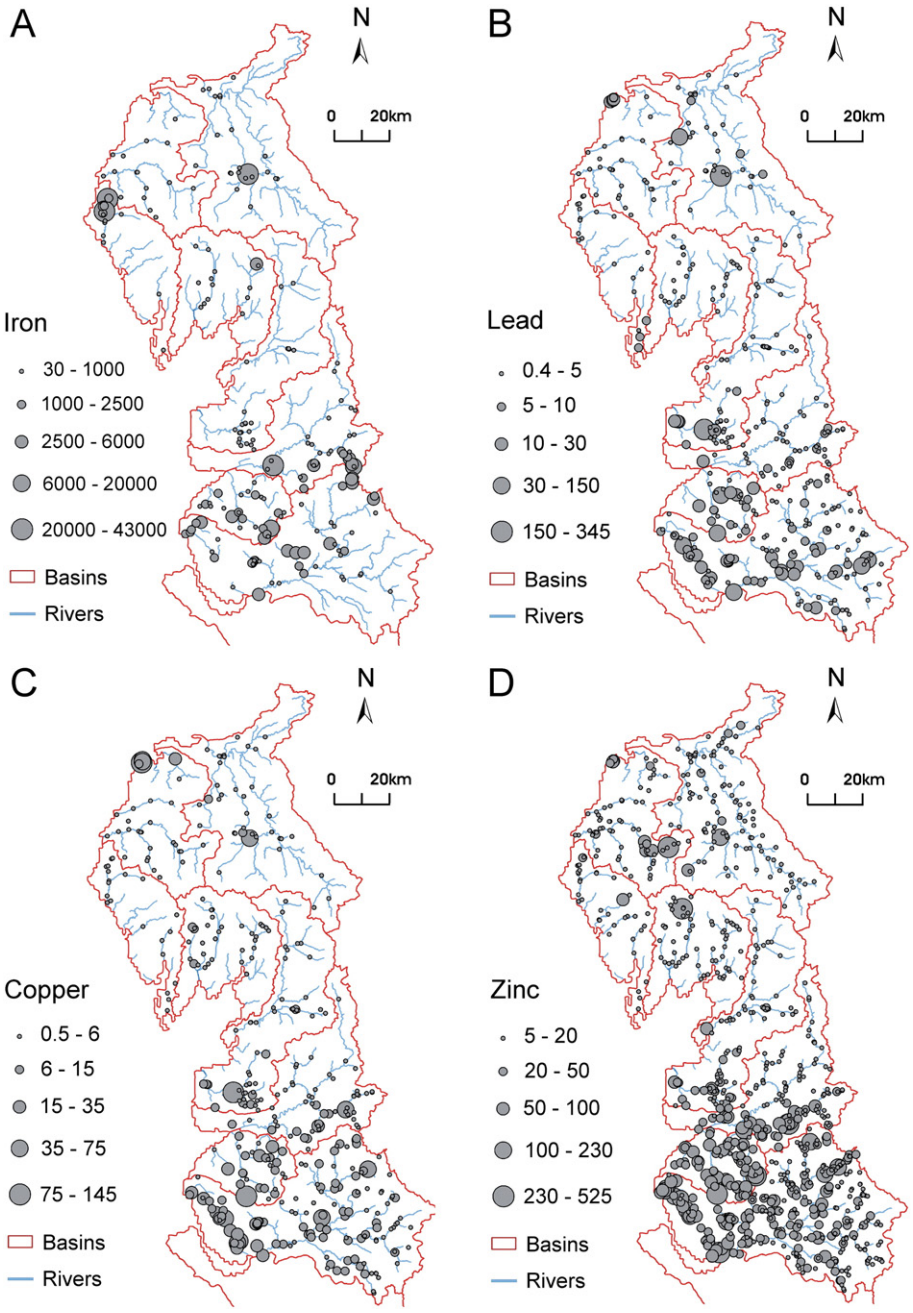


Fig. 5. Maps of average water chemistry across the North West. (A) Fe^{3+} ($\mu g\, l^{-1}$); (B) Pb^{2+} ($\mu g\, l^{-1}$); (C) Cu^{2+} ($\mu g\, l^{-1}$); (D) Zn^{2+} ($\mu g\, l^{-1}$).

average SO_4^{2-} concentration (Fig. 6A). There is also a divergent trend for Cu^{2+} with PO_4^{3-}-P (Fig. 6B). In both cases the relationship does not appear to be influenced by the density and proximity of discharge consents in the catchments. For those monitoring sites characterised by no or very few discharge consents, low to moderate NO_3^- concentrations are associated with low average PO_4^{3-} concentrations (Fig. 6C). For those Environment Agency monitoring sites with discharge consents within close proximity and a high density in the catchment there is a positive correlation between NO_3^--N and PO_4^{3-}-P (Fig. 6C).

4.4. Summer versus winter concentrations

At the majority of monitoring sites, summer pH, Ca^{2+}, Mg^{2+}, SO_4^{2-} concentrations are higher than those during the winter months (Fig. 7). Ammonium concentrations are generally elevated during winter months, with the exception of sites in Derwent, Wampool, Leven and Kent basins (Fig. 7E). For the majority of sites, NO_2^--N is elevated during the summer months (Fig. 7F). This is especially the case for the sites with highest NO_2^- concentrations (>0.1 mg-N l^{-1}). Winter months are characterised by high NO_3^- concentrations, with the exception of sites with the highest average concentrations (Fig. 7G). There is no clear pattern for sites with PO_4^{3-} concentrations <0.1 mg-P l^{-1} (Fig. 7H); some have elevated winter concentrations, others have elevated summer concentrations. However, for those sites with PO_4^{3-} concentrations >0.1 mg-P l^{-1}, elevated concentrations almost always occur during the summer months. Iron, Pb^{2+} and Cu^{2+} values are elevated in winter (Fig. 7I, J and K, respectively). Zinc concentrations are generally elevated during winter months except for those sites with average concentrations >100 μg-Zn l^{-1} (Fig. 7L).

Table 1
Bivariate Pearson correlations for average chemical concentrations.

	Acidity and base cations				Nutrients				Metals			
	pH	Ca^{2+}	Mg^{2+}	SO_4^{2-}	NH_4^+-N	NO_2^--N	NO_3^--N	PO_4^{3-}-P	Fe^{3+}	Pb^{2+}	Cu^{2+}	Zn^{2+}
pH	1.00											
Ca^{2+}	0.43	1.00										
Mg^{2+}	0.15	0.71	1.00									
SO_4^{2-}		0.59	0.71	1.00								
NH_4^+-N		0.22	0.18	0.28	1.00							
NO_2^--N	−0.08	0.37	0.28	0.26	0.47	1.00						
NO_3^--N		0.45	0.25		0.22	0.72	1.00					
PO_4^{3-}-P	−0.15	0.22	0.12		0.33	0.65	0.73	1.00				
Fe^{3+}				0.59					1.00			
Pb^{2+}		0.20	0.15	0.25		0.16				1.00		
Cu^{2+}		0.30	0.25	0.26		0.27	0.25	0.19		0.61	1.00	
Zn^{2+}		0.21	0.18	0.42	0.09	0.20	0.16	0.21		0.23	0.42	1.00

Correlations are only shown where $p<0.01$.

4.5. Relationships between water chemistry, catchment characteristics and other metrics

Significant ($p<0.01$) relationships were found between river water chemistry and catchment geomorphology, land cover, hydrology of soils type, baseflow index, geology, rainfall, atmospheric deposition (Table 2) and some of the consent metrics (Table 3). However, the strength and direction of the relationship varied between determinands. The key findings are summarised below.

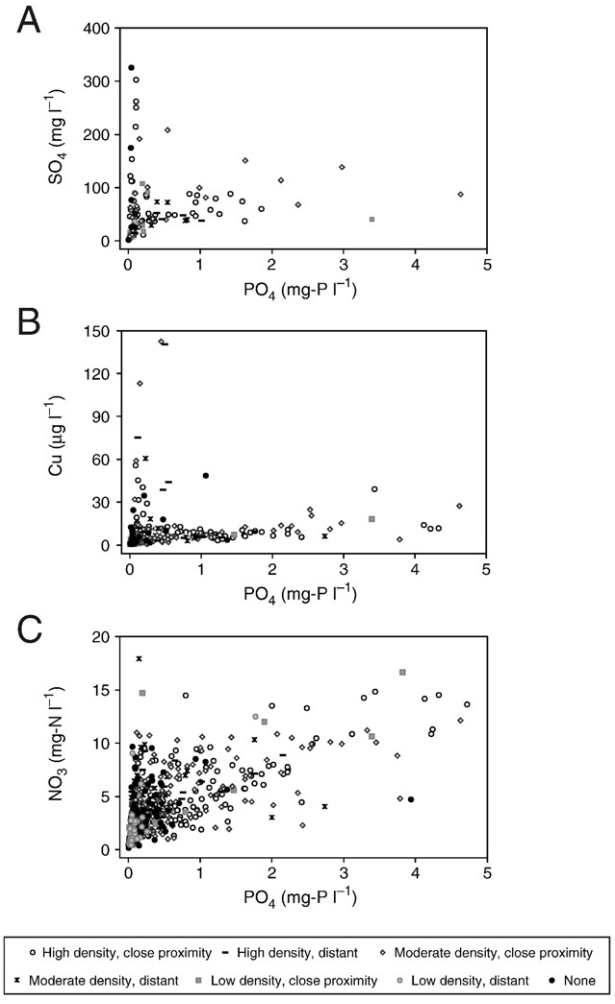


Fig. 6. Inter-determinand relationships for average chemical concentrations. (A) SO_4^{2-} (mg l^{-1}) and PO_4^{3-} (mg-P l^{-1}); (B) Cu^{2+} (µg l^{-1}) and PO_4^{3-} (mg-P l^{-1}); (C) NO_3^- (mg-N l^{-1}) and PO_4^{3-} (mg-P l^{-1}).

4.5.1. Acidity and base cations

There were good negative relationships between pH and average altitude and slope, percentage cover of bracken, acid grassland, the hydrology of peat soils, volcanic tuff, no superficial geology, and average annual rainfall. There was a statistically significant ($p<0.01$) positive relationship between pH and percentage cover of improved grassland and till, as well as between Mg^{2+} and percentage cover of arable, calcareous grassland, urban, gleyed soils, sandstone and mudstones, and till. There were also statistically significant ($p<0.01$) negative relationships between Mg^{2+} and average altitude and slope, percentage cover of acid grassland, heath, the hydrology of peat soils, no superficial geology, average annual rainfall and NH_3/NH_4 deposition. Sulphate was positively related to percentage cover of urban and coal measures (Table 2).

Table 3 show the relationships between average determinand concentration and discharge consent metrics (proximity and density). A weak, but significant ($p<0.01$), negative relationship was found between pH and the proximity of sites discharging sewage effluent from storm overflows. Weak positive relationships between the density of sites discharging sewage effluent from storm overflows and Mg^{2+} and SO_4^{2-} were also found. A weak positive relationship between SO_4^{2-} and density of sites discharging crude sewage effluent was found.

4.5.2. Nutrients

A statistically significant ($p<0.01$) positive relationship between NO_2^--N and percentage cover of sandstones and mudstones was found (Table 2). Statistically significant ($p<0.01$) negative relationships were obtained for NO_2^--N and average altitude and slope, and NH_3/NH_4 deposition, as well as positive relations between NO_3^--N and percentage cover of arable land and sandstones and mudstones. NO_3^- concentrations were negatively related to average altitude and slope ($p<0.01$). No strong relationships between PO_4^{3-}-P and catchment characteristics were obtained, although weak positive relations were found for percentage cover of arable and urban. The analysis revealed significant ($p<0.01$), but very weak relationships between average nutrient concentrations and the discharge consent metrics (Table 3).

In catchments where arable land is >50%, average NO_3^- concentrations generally exceed 5 mg-N l^{-1}, irrespective of the density or proximity of discharge consents (Fig. 8A). At monitoring sites with >1% arable land and >0.1% urban cover, average NO_3^- concentrations are highly variable. Those monitoring sites with no discharge consents within their catchments still have a wide range of NO_3^- concentrations when arable land is >1% (Fig. 8B). A similar pattern occurs between NO_3^--N and urban cover (Fig. 8C and D). The relationship between percentage arable land and average PO_4^{3-} concentrations is shown in Fig. 8E. Above 1% arable cover, PO_4^{3-} concentrations are highly

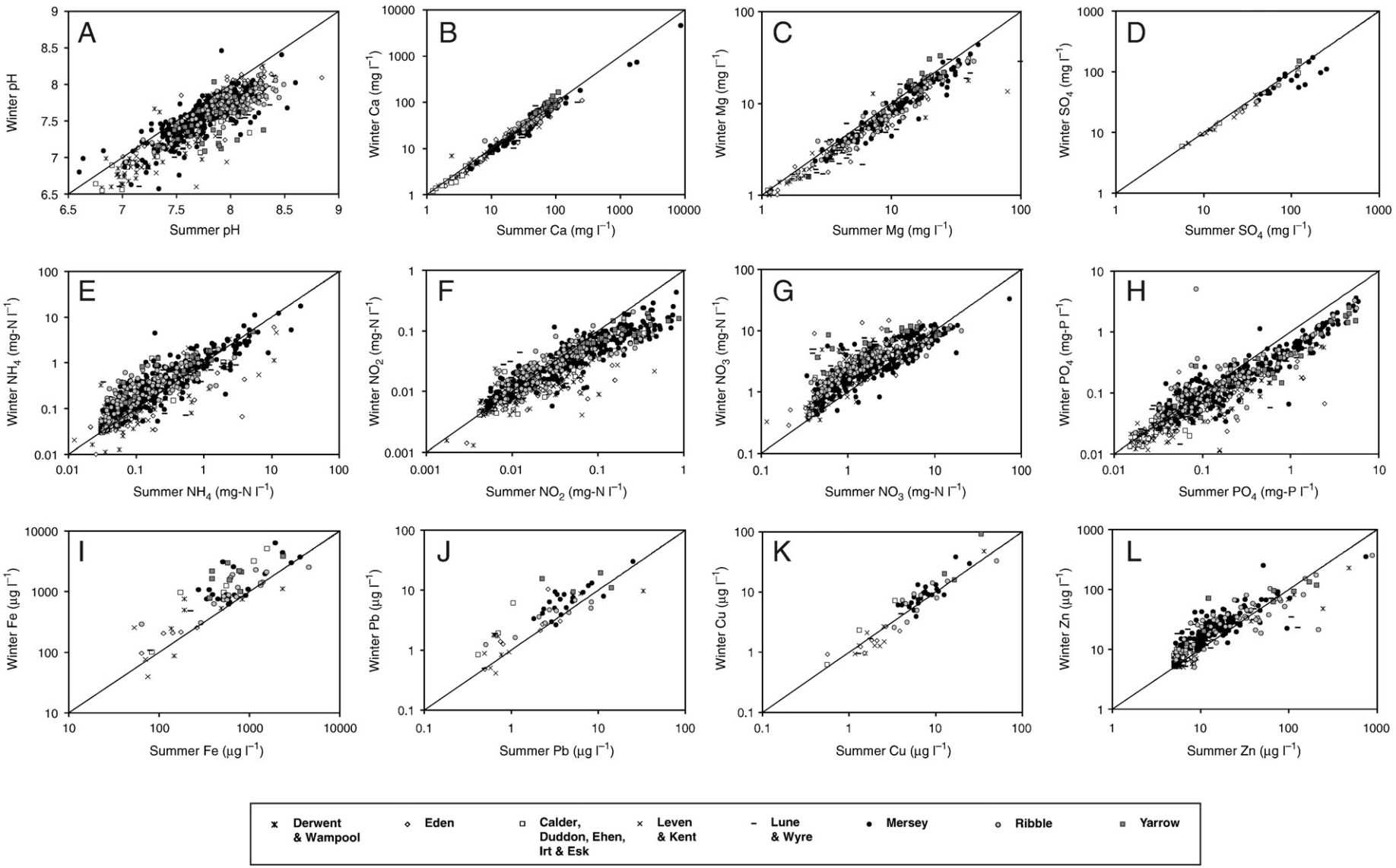


Fig. 7. Average summer vs average winter water chemistry. (A) pH; (B) Ca^{2+} (mg l^{-1}); (C) Mg^{2+} (mg l^{-1}); (D) SO_4^{2-} (mg l^{-1}); (E) NH_4^+ (mg-N l^{-1}); (F) NO_2^- (mg-N l^{-1}); (G) NO_3^- (mg-N l^{-1}); (H) PO_4^{3-} (mg-P l^{-1}); (I) Fe^{3+} (µg l^{-1}); (J) Pb^{2+} (µg l^{-1}); (K) Cu^{2+} (µg l^{-1}); (L) Zn^{2+} (µg l^{-1}).

Table 2
Bivariate Pearson correlations between average determinand concentration and a suite of catchment characteristics, rainfall and atmospheric deposition.

	Acidity and base cations				Nutrients				Metals			
	pH	*Ca^{2+}	Mg^{2+}	SO_4^{2-}	NH_4^+-N	NO_2^--N	NO_3^--N	PO_4^{3-}-P	Fe^{3+}	Pb^{2+}	Cu^{2+}	Zn^{2+}
DTM												
Area	0.12					0.23	0.11	0.13				
Aspect	−0.16								0.2			
Altitude	−0.42	−0.15	−0.75		−0.22	−0.44	−0.42	−0.3		−0.23	−0.49	−0.19
Slope	−0.56	−0.13	−0.66	−0.2	−0.18	−0.4	−0.38	−0.26			−0.44	−0.18
Land cover												
Arable	0.15	0.15	0.55	0.21	0.16	0.32	0.43	0.23			0.41	
Exposed rock	−0.14		−0.16				−0.15					
Bracken	−0.56		−0.38			−0.19	−0.21	−0.15				
Grassland — acid	−0.42		−0.52		−0.11	−0.22	−0.24	−0.17				
Grassland — calcareous	0.38		0.42									
Grassland — improved	0.42						0.2					
Grassland — neutral			−0.17		−0.1	−0.21	−0.24	−0.14				
Heath	−0.34		−0.4		−0.12	−0.23	−0.26	−0.14				
Urban	0.19		0.56	0.5	0.23	0.36	0.11	0.28		0.18	0.25	0.3
Water												
Wetland	−0.3		−0.26			−0.15	−0.2					
Woodland										0.28		
HOST												
Permeable soils	0.15		−0.27			−0.12				0.42		
Impermeable soils	−0.21		−0.27	−0.2		−0.14	−0.1					
Gleyed soils	0.39		0.68	0.23	0.13	0.26	0.14	0.15	0.19			
Peat soils	−0.43		−0.48		−0.1	−0.12	−0.13					
Base Flow Index	0.17		0.15		0.11	0.15	0.35	0.14		0.38	0.47	
Bedrock geology												
Coal Measures	0.13		0.24	0.46					0.2			
Conglomerates			−0.14									
Granites and lavas	−0.18		−0.17									
Limestone	0.28		−0.25	−0.23		−0.18		−0.12				
Shales	−0.13		−0.28			−0.19	−0.14	−0.13				
Siltstones			0.16	0.34	0.13	0.15						
Sandstones and mudstones	0.16	0.18	0.64	0.19	0.25	0.43	0.41	0.31		0.3	0.47	
Sandstones and shales			−0.3		−0.13	−0.15	−0.12					
Volcanic tuff	−0.57		−0.33			−0.16	−0.18	−0.14				
Superficial geology												
Alluvium												0.2
Peat	−0.18		−0.18									
Sands		0.23	0.24	0.27	0.18	0.17	0.23				0.5	
Till	0.47		0.48			0.19	0.12	0.13				
No superficial	−0.45	−0.13	−0.59		−0.16	−0.34	−0.26	−0.25				
Rainfall	−0.53		−0.41	−0.24	−0.11	−0.23	−0.31	−0.18		−0.26	−0.36	
Atmospheric deposition												
NH_3/NH_4	−0.28		−0.69		−0.21	−0.4	−0.39	−0.26			−0.37	
Ca + Mg							−0.13				−0.29	
SO_2/SO_4	−0.39		−0.38	0.19		−0.15	−0.34				0.13	
$NO_2/NO_3/HNO_3$	−0.25		−0.34			−0.14	−0.32				−0.35	

Correlations are only shown where $p<0.01$. * See discussion for an explanation of the poor relationships for Ca^{2+}.

variable, with average concentrations exceeding 5 mg-P l^{-1}, but also as low as 0.1 mg-P l^{-1}. However, the highest PO_4^{3-} concentrations are at sites characterised by high to moderate densities of discharge consents within the catchment and monitoring sites in close proximity to these consents (Fig. 8E). For those monitoring sites with no discharge consents within the catchment, PO_4^{3-} concentrations are much lower (Fig. 8F) although PO_4^{3-} concentrations broadly increase as the percentage of arable land increases. A similar pattern is seen for the relationship between percentage urban cover and average PO_4^{3-} concentrations (Fig. 8G and H).

4.5.3. Metals

For the metals, there were strong statistically significant ($p<0.01$) relationships between Pb^{2+} and percentage cover of permeable soils, Cu^{2+} and the percentage cover of arable land, sandstones and mudstones, and superficial sands, and also the baseflow index. There were strong statistically significant ($p<0.01$) negative relationships between Cu^{2+} and average altitude and slope (Table 2). A weak relationship between Fe^{3+} and the density of sites discharging mine effluent was found (Table 3). A statistically significant ($p<0.01$) positive relationship was found between Cu^{2+} and mine proximity (Table 3).

5. Discussion

The results of this study provide insight into the patterns and broad scale controls on river water chemistry across the North West and the utility of routine monitoring databases and spatial datasets for exploring regional water chemistry. The following discussion is structured into: acidity and base cations, nutrients and metals.

Table 3
Bivariate Pearson correlations between average determinand concentration and the metrics of proximity to discharge consent site and the density of discharge consent sites in a catchment.

	Acidity and base cations				Nutrients				Metals			
	pH	Ca^{2+}	Mg^{2+}	SO_4^{2-}	NH_4^+-N	NO_2^--N	NO_3^--N	PO_4^{3-}-N	Fe^{3+}	Pb^{2+}	Cu^{2+}	Zn^{2+}
Proximity												
Sewage — crude	−0.23		−0.31		−0.10	−0.22	−0.13	−0.17				
Sewage — storm	−0.41		−0.27	−0.19		−0.14		−0.13				
Sewage — treated	−0.32						−0.12	−0.15				
Trade — crude	−0.37	−0.12	−0.27		−0.10	−0.22	−0.17	−0.19		−0.16		
Trade — mine	−0.20				0.15	0.16	0.11	0.13			0.44	
Trade — treated	0.18		0.29	0.30		0.16		0.18				
Density												
Sewage — crude			0.36	0.40								
Sewage — storm			0.40	0.40								
Sewage — treated			0.35	0.21								
Trade — crude			0.13									0.17
Trade — mine									0.31			
Trade — treated			−0.20		−0.09	−0.13		−0.11				

Correlations are only shown where $p<0.01$.

5.1. Acidity and base cations

There is a clear link between river pH, catchment geology and land cover (Table 2). Igneous lithologies (granites, lavas and volcanic tuff; Fig. 1D) and land cover typified by vegetation associated with upland environments (e.g. bracken, heath and wetlands; Fig. 1C) tend to be characterised by acid waters. Catchments dominated by calcareous or improved grasslands (e.g. those in the Eden valley and the northern coastal plain of Cumbria; Fig. 1C) tend to be associated with higher pHs (Fig. 3A, Table 2). This is in agreement with other similar studies (e.g. Smart et al., 1998; Thornton and Dise, 1998). Results also suggest that the proximity of discharge consents in catchment may affect river water pH (Table 3), but it is difficult to disentangle the contributions from effluent inputs and ground water flow rich in base cations. A weak, but significant ($p<0.01$) relationship was found between river pH and the deposition of sulphur (Table 2). Although atmospheric sulphur deposition can have significant impacts on fluvial systems (Stoddard et al., 1999; Evans et al., 2001; Driscoll et al., 2003; Davies et al., 2005; Skjelkvale et al., 2005), it is likely that the relationship between river pH and the deposition of sulphur (Table 2) is due to the co-correlation of the presence of acidic upland soils (under wetland and heath land cover, Fig. 1B and C) and elevated sulphur deposition at high altitudes (Fowler et al., 1995).

Previous studies for a variety of UK catchments have found the distribution of Ca^{2+} concentrations in surface waters to depend on geology and the weathering of parent materials (Robson and Neal, 1997; Smart et al., 1998; Oguchi et al., 2000; Jarvie et al., 2002). Even though there are clear spatial differences in Ca^{2+} concentrations across the North West of England (Fig. 3B), very few linkages were found between the average Ca^{2+} concentration and the catchment characteristics investigated (Table 2). This is quite striking when compared to the clear linkages between Mg^{2+} and the geological/soil characteristics (Table 2). These results for Ca^{2+} can be explained by the influence of point source contributions. Three sites in the Mersey basin are characterised by very high summer Ca^{2+} concentrations (>1000 mg l^{-1}, Fig. 6B). Within 80 m of each of these monitoring sites crude effluent discharges are present. Although no strong relationships were found between Ca^{2+} and the discharge consent metrics, a weak significant ($p<0.01$) relationship was found between Ca^{2+} concentrations and the proximity of crude effluent sources (Table 3). When the three sites with excessive Ca^{2+} concentrations are removed from the analysis, the relationships between Ca^{2+} concentrations and catchment topography, geology and soils improve significantly and reflect the results of Mg^{2+}.

The highest Mg^{2+} concentrations occur in low elevation areas of the region (Fig. 3C) and have a similar distribution to that of Ca^{2+} (Table 1 and Fig. 3B). The results suggest that Mg^{2+} is mainly linked to diffuse geological sources, consistent with the work of Edmunds and Kinniburgh (1986). However, elevated Mg^{2+} concentrations are associated with agricultural activities (arable land and calcareous grassland), and urbanised areas (Table 2). Although agriculture could be a source of Mg^{2+}, the results also reveal that the density of sewage effluent discharges in a catchment could also play a role in determining stream water Mg^{2+} concentrations (Table 3). Robson and Neal (1997) and Jarvie et al. (2002) also suggest that urban areas and sewage effluent inputs are a major source of Mg^{2+} in surface waters for the industrial areas of the North East of England.

Although the sparse distribution of SO_4^{2-} data (Fig. 3D) make interpretation difficult, the main sources of SO_4^{2-} appear to be sewage effluent and bedrock geology (coal measures) (Table 2). Fig. 6A shows a clear separation in the relationship between SO_4^{2-} and PO_4^{3-}-P. Those sites with very high SO_4^{2-} values and low PO_4^{3-}-P values are characterised by either disused mines/collieries or areas of historical dumping of cinders and ashes (e.g. Irwell Mill in Bacup). The oxidation of sulphur-rich coal and leaching of SO_4^{2-} is a likely explanation for the elevated SO_4^{2-} values at these sites. The trend of moderate SO_4^{2-} concentrations and low to high PO_4^{3-}-P values (Fig. 6A) is probably indicative of areas receiving domestic sewage effluent. The link between point sources and PO_4^{3-}-P is examined in detail below. There is a weak, but significant ($p<0.01$), relationship between average SO_4^{2-} and the atmospheric deposition of sulphur (Table 2). The contribution of atmospheric sulphur to riverine SO_4^{2-} can be significant (Stoddard et al., 1999), but the relationship in the present data may be the result of co-correlation between mining areas in the uplands and elevated sulphur deposition at altitude. It is worth noting that headwater sites are under-represented in the EA dataset and as such this data may not be suitable for examining linkages between acid deposition and river water chemistry. Due to requirements of the Water Framework Directive (WFD), routine river water quality monitoring by the Environment Agency is changing. Monitoring using a river basin approach will replace the General Quality Assessment Scheme in 2009 (EA, 2009b). There will be more focus on monitoring and improving rivers in the context of tackling diffuse pollution, especially those at the upper end of the water quality scale. Despite the requirements of the WFD applying to upland headwaters, characterising the variability in these systems will be a major challenge (Temnerud and Bishop, 2005).

5.2. Nutrients

The results reveal that are no clear relationships between average NH_4^+ concentrations and the characteristics and metrics investigated

(Tables 2 and 3). Jarvie et al. (2002) and Davies and Neal (2004) found good positive correlations between average NH_4^+ concentrations and the proportion of urban cover in the industrial North East of England (the Humber basin). In the present investigation only a weak positive relationship was found between NH_4^+-N and percentage of urban cover (Table 2). Although it is evident that river NH_4^+ concentrations are, in part, influenced by human population in the North West region, the results indicate that NH_4^+ concentrations are controlled by other factors. The strong seasonality in NH_4^+ concentrations (Fig. 7E) suggests that terrestrially based nitrogen dynamics and within-stream processes and conditions are a major control on NH_4–N in surface waters in the North West. Increased biological activity during summer months and greater in-stream nitrification is a likely explanation for the seasonal trend in NH_4^+-N (Admiraal and Bottermans, 1989; Webster et al., 2003; Peterson et al., 2001; Ensign et al., 2006). Although sewage treatment works are a source of NH_4^+-N in surface waters (Admiraal and Bottermans, 1989; Jarvie et al., 1997), these inputs are highly sporadic and there will be variability in the attenuation of NH_4^+-N. This could mask relationships between stream NH_4^+-N and the suite of catchment characteristics investigated. Difference in nitrogen cycling between managed/populated areas and semi-natural environments may also explain the poor relationships between average NH_4^+-N and the catchment characteristics.

Previous studies have found a strong link between stream water NO_2^- concentrations and urban cover (e.g. Davies and Neal, 2004). Only a weak relationship between river NO_2^- concentrations and urban cover was found for the North West region (Table 2). Similarly, a weak, but significant relationship between NO_2^--N and arable land was obtained (Table 2). Although these results do not rule out urban and arable areas as sources of NO_2^--N to surface waters in the region, it is clear that other factors control river and stream NO_2^- concentrations. The difference between summer and winter NO_2^- concentrations (Fig. 7F) suggests that biogeochemical processes and conditions (e.g. temperature, moisture) are important controls on NO_2^--N in surface waters in the North West. Variability in these processes and conditions across the river network may explain the poor relationships between and the characteristics and metrics investigated (Tables 2 and 3).

The results of this study indicate that NO_3^--N is linked to both diffuse (agricultural) and point (sewage effluent) source inputs. The relationship between NO_3^--N and arable land (Table 2), and the elevated winter NO_3^- concentrations (Fig. 7G) points to diffuse agricultural sources (surface runoff from fields) being important contributors to riverine NO_3^--N. This is in agreement with other similar studies (e.g. Thornton and Dise, 1998; Ferrier et al., 2001; Jarvie et al., 2002; Davies and Neal, 2007; Neal et al., 2008a). Further evidence for the role of diffuse inputs is also seen in the scatter plots of NO_3^--N and the percentage of arable land (Fig. 8A and B). High NO_3^- concentrations occur in catchments characterised by high percentage cover of arable land, irrespective of whether discharge consents are present or not (Fig. 8A). However, the majority of sites with high average NO_3^- concentrations (>10 mg-N l^{-1}) are those with a high density and close proximity of discharge consents (Fig. 8A and B). Although no clear link was found between NO_3^--N and the discharge consent metrics (Table 3), the results clearly show that point sources are of equal, if not greater, importance in the delivery of NO_3^--N to North West rivers, especially in the southern part of the region. Davies and Neal (2004) and Meynendonckx et al. (2006) have also shown that point sources are important for river NO_3^- concentrations. No link was found between river and stream NO_3^- concentrations and nitrogen deposition (Table 2). This is probably related to the major influence of lowland NO_3^--N inputs to the river network and the poor representation of headwater catchments that receive high nitrogen deposition.

The weak relationships between average PO_4^{3-} concentrations and percentage arable land and urban cover (Table 2) indicates that land use alone cannot be used to explain the spatial distribution of PO_4^{3-} concentrations across the North West region. The highest PO_4^{3-} concentrations are at sites characterised by high to moderate densities of discharge consents within the catchment and monitoring sites in close proximity to these point sources (Table 3, Fig. 8E – H). Although the majority of these point sources are located in the populated areas of the North West, the results reveal that point sources in rural locations also have a significant effect on river PO_4^{3-} concentrations (Figs. 2D, 4D and 8G). The scatter plot of average summer and average winter PO_4^{3-} concentrations reveals that the highest PO_4^{3-} concentrations occur during summer months (Fig. 7H). This pattern is consistent with a lower dilution potential of summer baseflow for point (sewage effluent) sources (Jarvie et al., 2005, 2006, 2008; Bowes et al., 2008). At those sites with high PO_4^{3-} concentrations, there is also elevated NO_3^--N (Table 1 and Fig. 6C). The dominance of point source contributions (especially in the southern basins — Fig. 2B) appears to mask the influence of agriculture on stream water nutrient concentrations — especially PO_4^{3-}-P. This was also noted by Robson and Neal (1997) for Eastern UK rivers. With the ability to separate out those sites with the possibility of a strong point source contribution, the impact of agricultural runoff on stream water PO_4^{3-} concentrations can be fully evaluated. At EA monitoring sites with low intensity agriculture (<1%) and no known point sources, average PO_4^{3-} concentrations are low (<0.06 mg-P l^{-1}), with the exception of four sites which have slightly higher average PO_4^{3-} concentrations (Fig. 8F). These findings suggest that in areas with low intensity agriculture, PO_4^{3-}-P inputs from small unknown rural sources may be effectively regulated through in-stream processes, such as biological uptake or sediment binding (House, 2003; Jarvie et al., 2005, 2008).

At sites with no discharge consents, average PO_4^{3-} concentrations do increase with an increase in the percentage cover of arable land (Fig. 8F). However, at these sites, average PO_4^{3-} concentrations do not exceed 0.7 mg-P l^{-1}. At sites with significant sewage effluent sources in the catchment, average PO_4^{3-} concentrations are very high (>5 mg-P l^{-1}) (Fig. 8E). The relations between discharge consent metrics and chemical determinands reveals that average PO_4^{3-}-P concentrations are elevated at EA monitoring sites with sewage effluent sources in close proximity, although the relation is weak (Table 3). Phosphorus is highly reactive and within-river losses through mechanisms such as sediment binding (House, 2003; Withers and Jarvie, 2008), may explain the poor linear relations between PO_4^{3-}-P and the consent site metrics.

Within the context of the WFD, diffuse nutrient inputs to aquatic systems are regarded as a major contributor to eutrophication of surface waters (Neal and Heathwaite, 2005; Davy et al., 2008). Point sources have historically been the main cause of water quality problems in England. The results of this study demonstrate that impacts from urban and rural point sources remain a serious problem in the context of surface water nutrient concentrations. According to the Environment Agency General Quality Assessment (GQA) scheme limits for nutrients, average NO_3^- concentrations across the region are 'moderate' (<30 mg-N l^{-1}) or better. However, 44% of sites in the North West have average PO_4^{3-} concentrations >0.1 mg-P l^{-1} and 8% of sites have 'excessively high' average concentrations (>1 mg-P l^{-1}). Many of the sites with elevated PO_4^{3-} concentrations are characterised by both close proximity to upstream point sources and high point source densities within the catchment. The introduction of the Catchment Sensitive Faming initiative in England (DEFRA, 2004) and other European policies to reduce diffuse nutrient inputs and improve water quality may be unsuccessful without sufficient attention to point sources. Although this may not apply to all regions, the results from North West England presented here demonstrate that both point and diffuse sources problems need to be tackled, especially if the objectives of 'good ecological status' and 'good chemical status' are to be met.

5.3. Metals

Metals preferentially associate with the sediment phase in river systems (Salomons and Forstner, 1984; Horowitz, 1995). Fine colloids

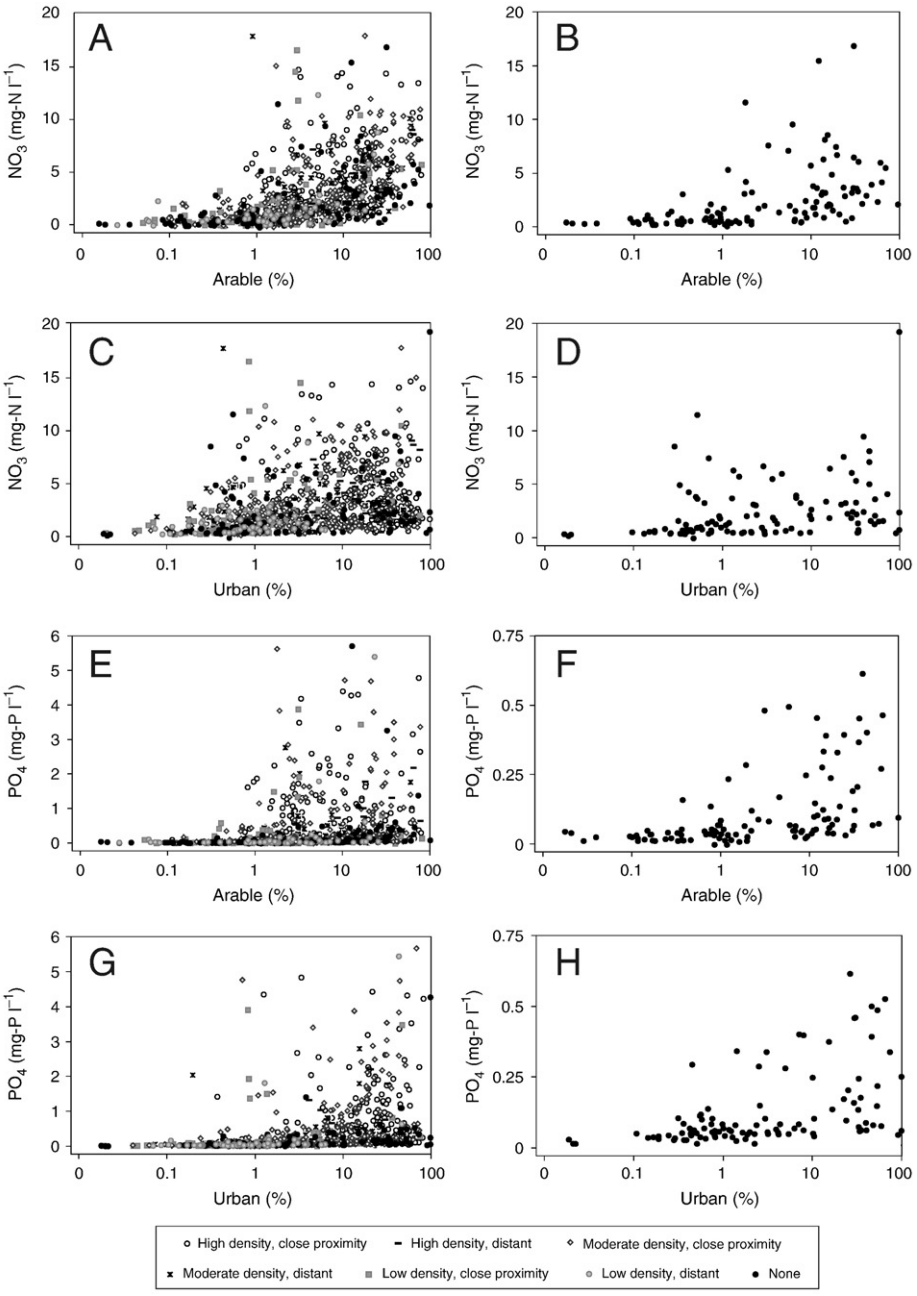


Fig. 8. Relationships between the average concentration of NO_3^- (mg-N l^{-1}) and the percentage cover of arable (A and B) and urban (C and D). Relationships between the average concentration of PO_4^{3-} (mg-P l^{-1}) and the percentage cover of arable (E and F) and urban (G and H). Panels A, C, E and G show monitoring sites with and without discharge consents. Panels B, D, F and H only show those monitoring sites with no discharge consents. Note the difference in scale for panels E to H.

and micro particulates are also important for metal transport in rivers (Neal et al., 1997, 2008b). Since the EA measure water chemistry on unfiltered samples, it is likely that suspended particulates, micro particulates and fine colloids will be present. As particulate transport in fluvial systems is strongly controlled by flow (Walling, 1977), there is greater potential for metal transport in rivers during high discharge conditions. In this study, the metals (Fe^{3+}, Pb^{2+}, Cu^{2+} and Zn^{2+}) are generally elevated during winter months (Fig. 7I – L). This is probably due to higher concentrations of particulates and colloids in the water samples during these conditions. Although this pattern suggests that that metals are derived from diffuse sources, as sediments and bound metals are mobilised by erosion, point sources cannot be ruled out if inputs occur during high flow. Even though there is no clear link between Fe^{3+} and the catchment characteristics investigated (Table 2), there is a link between the density of mine effluent discharges and river Fe^{3+} concentrations (Table 3). The results suggest that mining effluents are a significant source of Fe^{3+} in North West rivers. Mining areas are known to be significant sources of a wide range of metals (Salomons and Forstner, 1984). Oguchi et al. (2000) also suggested that mine drainage is a source of Fe^{3+} in river waters in the Humber catchment.

The results reveal that Pb^{2+} is linked to the hydrology of soil type (permeable soils), and the baseflow index (Table 2). Catchments with a significant groundwater contribution to total flow (high BFI) and free draining soils have elevated Pb^{2+} concentrations in stream water. These catchments tend to be dominated by carboniferous lithologies (particularly limestone and sandstones and shales). Previous phases of ore mining and associated pollution of soils and floodplain sediments at these locations may be one explanation for the elevated stream Pb^{2+} concentrations (Robson and Neal, 1997). Catchments with high a BFI and free draining soils in the North West also have significant levels of

urban cover. Runoff from urban areas can deliver Pb^{2+} and other metals to the fluvial system (Sansalone and Buchberger, 1997; Sorme and Lagerkvist, 2002). Although the results reveal a weak link between Pb^{2+} and urban cover (Table 2), these areas may still be significant sources of Pb^{2+}. Due to the strong binding of Pb^{2+} to fluvial sediments (Salomons and Forstner, 1984), overbank sediment losses and in-stream retention (in channel bed sediments) may be one explanation for the poor relationship between Pb^{2+} and urban cover.

High Cu^{2+} concentrations in river water occur in catchments characterised by a high percentage cover of arable land and this indicates a dominant agricultural source (Table 2). These catchments are free draining since they have a high BFI and this links to superficial and bedrock geology dominated by sandstone (Table 2). The application of pesticides, sewage sludge and livestock manure can lead to significant inputs of Cu^{2+} to agricultural land (Nicholson et al., 2003). Water erosion on agricultural soils can be a major mechanism for the delivery Cu^{2+} and other metals to fluvial systems (Quinton and Catt, 2007). The results of the present study also suggest that point sources can also be important Cu^{2+} sources. Fig. 6B revealed a clear separation in structure for the relationship between Cu^{2+} and PO_4^{3-}-P. The sites with very high Cu^{2+} values (>30 μg l^{-1}) and low PO_4^{3-}-P values are associated with discharges from colliery spoil, airfields and chemical works.

A link was found between Zn^{2+} and the percentage of urban cover, although the relationship was weak (Table 2). Sediment–solution interactions and the potential for non-conservative behaviour of Zn^{2+} in the water column may be one explanation for the poor relationship between Zn^{2+} and urban cover. The very high levels of Zn^{2+} almost always occur during summer months (Fig. 7L), when compared to winter months, suggesting a different source of waters rich in Zn^{2+}. A link was found between Zn^{2+} and density of crude trade effluent discharges (Table 3). The elevated summer concentrations may be consistent with a lower dilution potential of summer baseflow for these trade effluent sources. The results suggest that both urban and industrial areas are sources of Zn^{2+} in North West surface waters. Interestingly, Jarvie et al. (2002) found Zn^{2+} to be linked to arable in the Humber catchment. The dominance of point source contributions in the south of the North West region (Fig. 2B) may be masking the influence of agriculture on stream Zn^{2+} concentrations.

6. Conclusions

This study provides a spatial and seasonal assessment of the water chemistry across the North West region using an extensive dataset collected by the Environment Agency (EA) of England and Wales. The linkages between river water chemistry and catchment characteristics, rainfall, deposition chemistry and discharge consent site metrics have been explored using GIS mapping and analysis. The results reveal that human activities in the region have a significant imprint on river water chemistry. Orthophosphate concentrations are mainly linked to point sources. Nitrate is linked to agriculture, but point source inputs are also significant in the southern part of the region, particularly in south Lancashire, Greater Manchester and Merseyside. The importance of nutrient inputs from point sources in the North West is critical in relation to effective catchment management, especially in the context of meeting the goals of the WFD. Recently implemented WFD monitoring programmes in the North West reveals that 30% of surface waters are currently at 'good ecological status' (EA, 2009a). Although improvements in water quality are expected by 2015, a significant set of pressures remain, including tackling diverse nutrient sources and modified/artificial waterbodies.

River water chemistry (especially N and P concentrations) is highly variable due to a complex set of interacting hydrological and biogeochemical processes. Even though the Environment Agency sampling campaign (and other similar monitoring networks) may not capture all the variability in water chemistry dynamics, the GIS approach used in this study has demonstrated that this resource is great utility for identifying patterns and controls on river water chemistry. These data are a national asset, but have not been fully exploited for the analysis of water chemistry dynamics. The addition of flow data and soil physico-chemical data would strengthen the value in terms of linking water quality process and change to catchment hydrology and biogeochemistry. This provides a challenge in relation to free accessibility of data and methods of combining different data sets. Integration of GIS approaches and extensive monitoring data provides a base for underpinning environmental management at local, regional, national and international scales.

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